TASK 1

Development • Cybersecurity • Web3

A beginner-friendly practical task designed to build fundamentals through implementation, documentation, and demonstration.

1. Development — To-Do List

Objective: Build a functional To-Do List application. Members may choose either a frontend-only approach or a full-stack approach.

Tech Stack

- **Frontend:** HTML, CSS and JavaScript (or React, if preferred)
- **Optional backend:** Node.js + Express or any suitable backend
- **Database:** MongoDB, MySQL, PostgreSQL, or another suitable database

Requirements

- Add a task
- Edit a task
- Delete a task
- Mark/unmark a task as completed
- Display pending and completed tasks clearly
- **Frontend-only:** persist tasks using browser localStorage
- **Full-stack:** store tasks in a database and retrieve them through the backend

Optional Enhancements

- Task filters — All / Active / Completed
- Clear completed tasks
- Due dates or priorities
- Dark mode
- Responsive/mobile-friendly UI

Deliverables

- Project folder named after the member
- Source code
- README explaining the approach and setup
- 2–5 minute screen recording demonstrating the application

2. Cybersecurity — Basic Web Recon

Objective: Learn the basics of web reconnaissance and identify simple security observations using an intentionally vulnerable or authorized training target.

Tasks

- Choose an authorized training target (for example, a suitable TryHackMe room).
- Perform basic Nmap reconnaissance to identify open ports and services.
- Inspect the target website using browser Developer Tools.
- Check HTTP response headers, cookies, robots.txt and visible/interesting endpoints.
- Document at least **3 security observations** and explain why each may matter.
- For each observation, suggest a basic remediation.

Deliverables

- Folder named after the member
- Nmap/recon output and screenshots where relevant
- README containing commands used, observations, evidence and suggested remediation

Important: Only scan systems you own or have explicit permission to test. Use training platforms and intentionally vulnerable targets for this task.

3. Web3 — Simple Smart Contract

Objective: Write and interact with a very simple smart contract. The goal is to understand the basic idea of a blockchain transaction and contract interaction.

Task

- Create a simple Solidity smart contract that stores a number or a short message.
- Add a function to update the stored value.
- Add a function to read the stored value.
- Deploy and test the contract using Remix with a local/test environment.
- Interact with the contract and show the value changing.

Optional

- Connect the contract to MetaMask.
- Deploy it to a testnet.
- Build a very small HTML/JavaScript interface for reading and updating the value.

Deliverables

- Project folder named after the member
- Solidity contract
- README explaining the contract and how it was tested/deployed
- Screenshots or a short demo showing the contract working

Submission Checklist

Domain	Task	Minimum Deliverable

Development	To-Do List	Working app + README + demo
Cybersecurity	Basic Web Recon	Recon evidence + security report
Web3	Simple Smart Contract	Solidity contract + README + demo

Focus on understanding and implementation rather than using tutorials or generated code without understanding it.